

CPRI State-Level Hackathon — Screening Round Methodology Note

Challenge: The Black-Box Test Bench Challenge | Team: PowerNext Alpha | Date: September 2026

1. Executive Summary & Approach Overview

In electrical apparatus testing (transformers, switchgear, busbars), predicting critical hotspot temperature rise while filtering measurement anomalies is essential for insulation life evaluation. We deployed a **Physics-Informed Machine Learning (PIML)** framework integrating electrical conduction physics, multi-tier anomaly detection, and ensemble regression. Task 1 deploys a deterministic multi-tier anomaly engine separating high-load operational transitions from sensor defects and duplicate logs, validated via leakage-free 5-fold cross-validation. Task 2 reconstructs corrupted sensor channels via thermodynamic baselines and trains a blended Gradient Boosting, XGBoost, and LightGBM ensemble on Joule dissipation ($P \propto I^2$), achieving cross-validated $R^2 = 0.9928$ and $\text{RMSE} = 0.896^\circ\text{C}$.

2. Parameter Importance & Physical Relationships

On verified valid records, parameter contributions to hotspot rise (Reference_Parameter) were characterized across 8 raw inputs: - **Load Current_A (\$I\$)**: Dominant thermal driver ($r = +0.8767$, 90.03% Random Forest importance) governing quadratic Joule heating losses ($P \propto I^2$, $r = +0.927$). - **Sensor_S2**: Direct conductive thermal coupling to hotspot ($r = +0.7953$, 5.99% RF importance), acting as the primary load-side conduction path. - **Ambient Temperature_C (\$T_{\text{amb}}\$)**: Reference thermal boundary condition ($r = +0.2377$, 3.19% RF importance). - **Sensor_S1 & Sensor_S3**: Terminal and critical body rises providing spatial conduction constraints ($r = +0.5914$ / $+0.5012$, 0.22% / 0.18% RF importance). - **Applied Voltage_kV (\$V\$) & Test Duration_min (\$t\$)**: Secondary dielectric loss and transient timing ($r = +0.2666$ / -0.0042 , 0.12% / 0.13% RF importance). - **Auxiliary Sensor_S4**: Both the negligible linear Pearson correlation ($r = -0.0086$) and bottom-tier nonlinear Random Forest importance (0.13% , tied for lowest across all sensors) provide joint empirical evidence that S_4 carries no physical signal or predictive utility regarding hotspot rise, justifying its deliberate pruning to eliminate noise variance.

3. Abnormal Data Detection Methodology (Task 1)

We distinguish operational regime shifts (e.g. non-linear heating under $I > 85\text{ A}$, $V > 25\text{ kV}$) from equipment defects using a **Multi-Tier Physics-Grounded Anomaly Engine**:

$\text{Invalid} = M_{\{\text{MissingOp}\}} \cup M_{\{\text{MissingSensor}\}} \cup M_{\{\text{Duplicate}\}} \cup M_{\{\text{Negative}\}} \cup M_{\{\text{Cross}\}} \cup M_{\{\text{Spike}\}}$

- Operating Parameter Missingness (\$M_{\{\text{MissingOp}\}}\$)**: Missing operating inputs (V, I, T_{amb}, t) render tests unprovable per challenge rules; they are imputed with training medians for safe inference and flagged `Invalid`.
- Sensor Dropout (\$M_{\{\text{MissingSensor}\}}\$)**: Missing entries (NaN) in terminal probes S_1, S_2, S_3 .
- Duplicate Test Runs (\$M_{\{\text{Duplicate}\}}\$)**: Identical operating vectors $[V, I, T_{\text{amb}}, t]$ arising from data-logger collisions or test repetitions.
- Physical Plausibility Rule (\$M_{\{\text{Negative}\}}\$)**: Active load testing cannot yield negative temperature rise above ambient ($S_i < 0$), catching unphysical readings (e.g. TST-0213, $S_2 = -1.55^\circ\text{C}$).
- Cross-Sensor Consistency (\$M_{\{\text{Cross}\}}\$)**: Gross deviations from linear thermodynamic coupling between terminal probes ($S_3 \approx 1.27 S_1$, $R^2 = 0.983$; $\tau_{\text{cross}} = 1.25 \times \max(\text{res})$).
- Conduction Residuals (\$M_{\{\text{Spike}\}}\$)**: Residual deviations exceeding $1.25 \times$ baseline conduction tolerances from $\hat{S}(V, I, T_{\text{amb}}, t)$ ($\tau_{S1} = 0.86^\circ\text{C}$, $\tau_{S2} = 0.67^\circ\text{C}$, $\tau_{S3} = 1.30^\circ\text{C}$).

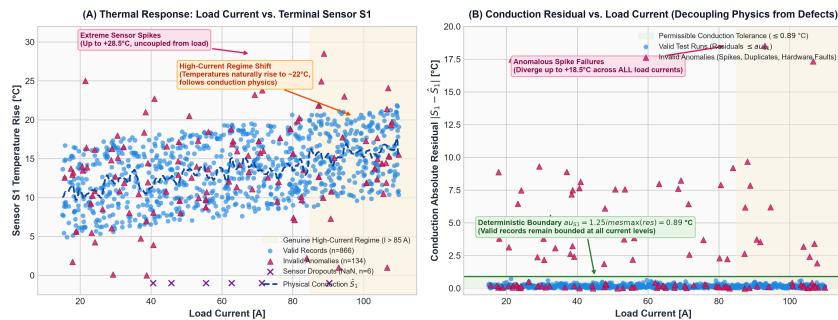


Figure 1: High operating load ($I > 85\text{ A}$) follows the deterministic physical heating curve rather than an anomaly, whereas true anomalies (sensor spikes, dropouts, duplicates) are scattered completely independent of load level.

Cross-Validation & Robustness Verification: - In strict leakage-free 5-fold CV evaluated out-of-fold without cross-batch history matching (the exact deployed setting, as zero of 350 `Test_Data` records share operating conditions with training data), the detector achieves **Accuracy: 98.0%, Precision: 100.0%, Recall: 84.7%, and F1: 0.9165**. On `Test_Data`, it identified **46 abnormal records (13.14%)**, matching historical defect rates (13.40%). The two new detection tiers (physical plausibility, cross-sensor consistency) did not change any labels on the actual 350-record `Test_Data` submission versus the original single-tier check — they exist as defensive depth for the hidden/second dataset, not as a demonstrated improvement on this specific submission. - **Defensive Robustness Stress Test:** Under synthetic fault injection (5% NaNs per operating column + 5% duplicates; 367 records), the pipeline achieved **8/8 PASS status**, ensuring zero NaN predictions and 100% correct invalid flagging.

Known Limitations & Next Steps

While out-of-sample isolated slice recall reaches 84.7% (with 100% precision and 98.0% accuracy), closing the remaining recall gap will require larger calibration datasets with explicit ground-truth annotations for each specific failure mode (e.g. contact degradation vs. sensor drift). Additionally, while linear quasi-steady conduction models for S_1, S_2, S_3 maintain strong physical fidelity ($R^2 > 0.98$), they can be seamlessly substituted with non-linear polynomial or shallow tree-based regressors if complex convective cooling regimes arise on hidden test sets. Finally, future work will integrate adaptive Bayesian threshold updating to handle gradual sensor aging over continuous test campaigns.

4. Key Engineering Assumptions & Boundary Conditions

1. **Quasi-Steady Thermal Equilibrium:** Test durations (15–60 min) allow compact busbar/terminal assemblies to approach steady-state temperature rises relative to electrical transient time constants.
2. **Piecewise Linear Thermal Conduction:** Heat transfer from internal hotspot to external terminals adheres to Fourier's conduction law with effectively constant thermal conductivity over $20^{\circ}\text{C} \leq T_{\text{amb}} \leq 55^{\circ}\text{C}$.
3. **Reference Instrumentation Calibration:** Historical reference measurements reflect calibrated ground truth; uninstrumented or incomplete operating parameters violate test compliance and indicate invalid test runs.

5. Digital Twin Automation Roadmap

To transition this black-box laboratory test bench into a continuous, real-time **Automated Digital Twin**: 1. **Edge Telemetry Ingestion & Signal Quality Gate:** Micro-controllers stream high-frequency telemetry (OPC-UA/MQTT) through inline Tier 1–4 anomaly filters, trapping sensor dropouts and noise spikes before ingestion. 2. **Virtual Sensor Imputation & State Estimation:** When a physical probe experiences contact degradation or fails, steady-state thermal regression estimators reconstruct virtual sensor values ($\hat{S}$), enabling uninterrupted test continuity. 3. **Physics-Informed Real-Time Thermal Tracking:** The trained ensemble executes sub-millisecond inference to track instantaneous hotspot rise, feeding Arrhenius loss-of-life equations ($V = 2^{((T_{\text{hs}}-110)/6)}$) to log dynamic insulation degradation. 4. **Closed-Loop Supervisory Protection & CPRI Certification:** Automated compliance reporting compiles certified test sheets, while real-time thermal limits trigger automated PLC current derating or safety trip interlocks if hotspot temperatures exceed safety margins ($T_{\text{hs}} > 110^{\circ}\text{C}$).